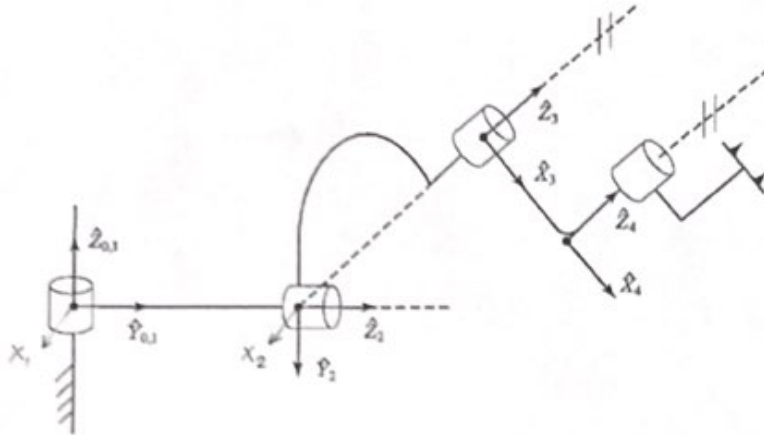


Problem 1 (40 pts)

A 4R manipulator is shown in Figure 1. The nonzero link parameters are $\alpha_1 = -90^\circ$, $d_2 = 1$, $\alpha_2 = 45^\circ$, $d_3 = 1$, and $a_3 = 1$. The mechanism is pictured in the configuration corresponding to $\theta = [0, 0, 90^\circ, 0]^T$.


 Figure 1: A 4R manipulator shown in the position $\theta = [0, 0, 90^\circ, 0]^T$.

1. Do the forward and inverse kinematics given ${}^0P_{4ORG} = [x, y, z]^T$. (15 pts + 15 pts)
2. Find all values of $[\theta_1, \theta_2, \theta_3]$ such that ${}^0P_{4ORG} = [0, 1, \sqrt{2}]^T$. (5 pts)
3. Find all values of $[\theta_1, \theta_2, \theta_3]$ such that ${}^0P_{4ORG} = [\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0]^T$. (5 pts)

$$\begin{array}{c|c|c|c|c}
 i & \alpha_{i-1} & a_{i-1} & d_i & \theta_i \\
 \hline
 1 & 0 & 0 & 0 & \theta_1 \\
 2 & -90^\circ & 0 & 1 & \theta_2 \\
 3 & 45^\circ & 0 & 1 & \theta_3 \\
 4 & 0 & 1 & 0 & \theta_4
 \end{array}
 \Rightarrow {}^0T = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ -\sin\theta_2 & -\cos\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 & 0 \\ \frac{\sqrt{2}}{2}\sin\theta_3 & \frac{\sqrt{2}}{2}\cos\theta_3 & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2}\sin\theta_3 & \frac{\sqrt{2}}{2}\cos\theta_3 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow x^2 + y^2 + z^2 = 3 + \sqrt{2} + \sqrt{2} \sin\theta_3 \Rightarrow \theta_3 = \arctan\left(\frac{x^2 + y^2 + z^2 - 3 - \sqrt{2}}{\sqrt{2}}\right), \pm \sqrt{1 - s_3^2}$$

$$\Rightarrow \underbrace{\frac{\sqrt{2}}{2}(1 - \sin\theta_3)}_m \cos\theta_2 - \underbrace{\cos\theta_3 \sin\theta_2}_n = z \Rightarrow \theta_2 = \arctan\left(\pm \sqrt{m^2 + n^2 - z^2}, z\right) + \arctan\left(\frac{A}{B}\right)$$

$$\begin{aligned}
 \Rightarrow \theta_1 &= \arctan\left(\frac{ay - xe, xf - cy}{ax - cy}\right) \\
 &= \arctan\left(\frac{ay + cx, ax - cy}{ax - cy}\right)
 \end{aligned}$$

$$\begin{aligned}
 2. \quad \begin{cases} x=0 \\ y=1 \\ z=\sqrt{2} \end{cases} &\Rightarrow \theta_3 = \operatorname{atan2}(-1, 0) = \underline{-90^\circ} \Rightarrow \begin{cases} m = \sqrt{2} \\ n = 0 \end{cases} \\
 &\Rightarrow \theta_2 = \operatorname{atan2}(0, \sqrt{2}) + \operatorname{atan2}(0, \sqrt{2}) = \underline{0} \Rightarrow \begin{cases} \alpha = 0 \\ c = -1 \end{cases} \\
 &\Rightarrow \theta_1 = \operatorname{atan2}(0, 1) = \underline{0}
 \end{aligned}$$

$$3. \quad \begin{cases} x = \frac{\sqrt{2}}{2} \\ y = \frac{\sqrt{2}}{2} \\ z = 0 \end{cases} \Rightarrow \sin \theta_3 = \frac{x^2 + y^2 + z^2 - 3}{\sqrt{2}} = \frac{1 + 1 + 0 - 3}{\sqrt{2}} = \frac{-1}{\sqrt{2}} < -1 \Rightarrow \theta_3 \in \emptyset$$

Problem 2 (30 pts)

Imagine an arm like PUMA 560, except that joint 3 is replaced with a prismatic joint. Assume the prismatic joint slides along the direction of $\hat{Z}_3$ shown in Figure 2. However, there is still an offset equivalent to d_3 to be accounted for. The schematic of the robot wrist is shown in Figure 3. The origin of frame 4, 5 and 6 are all at the same point.

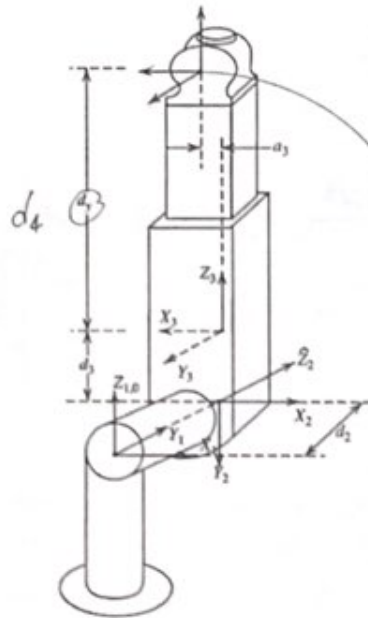


Figure 2: Frame assignments of the manipulator.

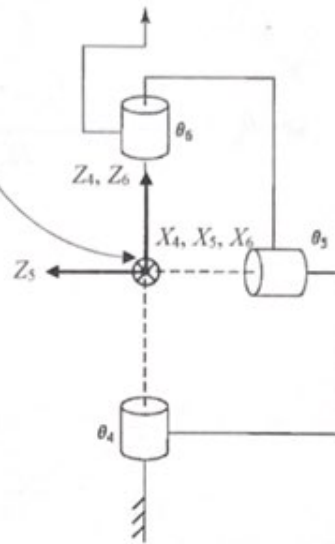


Figure 3: Wrist schematic.

1. Complete the DH table. (10 pts)

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	-90°	0	d_2	θ_2
3	90°	0	d_3	180°
4	0	a_3	d_4	θ_4
5	-90°	0	0	θ_5
6	90°	0	0	θ_6

2. Write out all the individual transformation matrices between joints and describe how to calculate the overall transformation matrix from frame 0 to frame 6. (You do **NOT** need to calculate 0_6T ; 10 pts)

3. Given ${}^4_6T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, solve for θ_5 and θ_6 symbolically. (10 pts)

$$\begin{aligned}
 2. \quad {}^0_1T &= \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^1_2T = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ -\sin\theta_2 & -\cos\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^2_3T = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -d_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 {}^3_4T &= \begin{bmatrix} \cos\theta_4 & -\sin\theta_4 & 0 & a_3 \\ \sin\theta_4 & \cos\theta_4 & 0 & 0 \\ 0 & 0 & 1 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^4_5T = \begin{bmatrix} \cos\theta_5 & -\sin\theta_5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sin\theta_5 & -\cos\theta_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^5_6T = \begin{bmatrix} \cos\theta_6 & -\sin\theta_6 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ \sin\theta_6 & \cos\theta_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

$$\Rightarrow {}^0_6T = {}^0_1T {}^1_2T {}^2_3T {}^3_4T {}^4_5T {}^5_6T$$

$$3. \quad {}^4_6T = {}^4_5T {}^5_6T = \begin{bmatrix} \cos\theta_5 \cos\theta_6 & -\cos\theta_5 \sin\theta_6 & \sin\theta_5 & 0 \\ \sin\theta_6 & \cos\theta_6 & 0 & 0 \\ -\cos\theta_5 \sin\theta_6 & \sin\theta_5 \sin\theta_6 & \cos\theta_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{cases} \theta_5 = \text{atan2}(r_{13}, r_{33}) \\ \theta_6 = \text{atan2}(r_{21}, r_{22}) \end{cases}$$

Problem 3 (15 pts)

Sketch the **dexterous** workspace of the RPR planar robot. All revolute joints can rotate 360° .

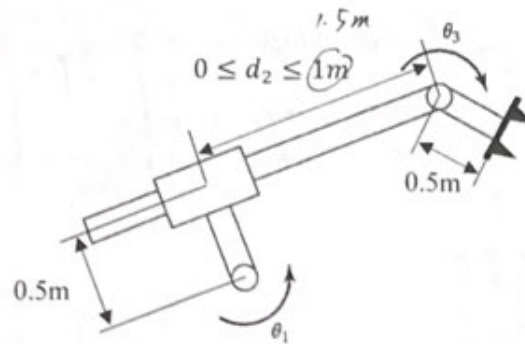
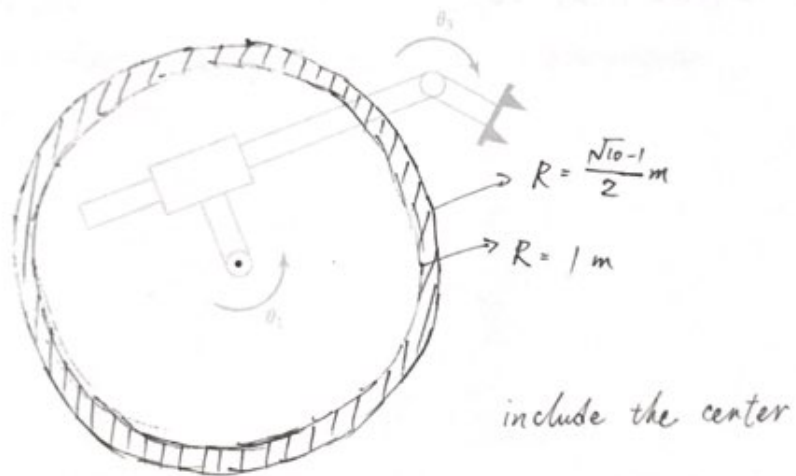


Figure 4: RPR planar robot.



Problem 4 (15 pts)

In Figure 5, two 3R mechanisms are pictured. In both cases, the three axes intersect at a point. The mechanism of (a) has link twists of magnitude 90 degrees. The mechanism of (b) has one twist of ϕ ($< 90^\circ$) in magnitude and the other of $180 - \phi$ in magnitude. The mechanism of (a) can be seen to be in correspondence with Z-Y-Z Euler angles, and therefore we know that it suffices to orient link 3 (with arrow in figure) arbitrarily with respect to the link 0. In other words, the arrow can point to the intersection in any direction. Because ϕ is not equal to 90 degrees, it turns out that the other mechanism cannot orient link 3 arbitrarily.

Describe or sketch the set of orientations that are **unattainable** with the mechanism of (b). Note that we assume that all joints can turn 360 degrees (i.e. no limits) and the links may pass through each other if need be (i.e., workspace not limited by self-collisions).

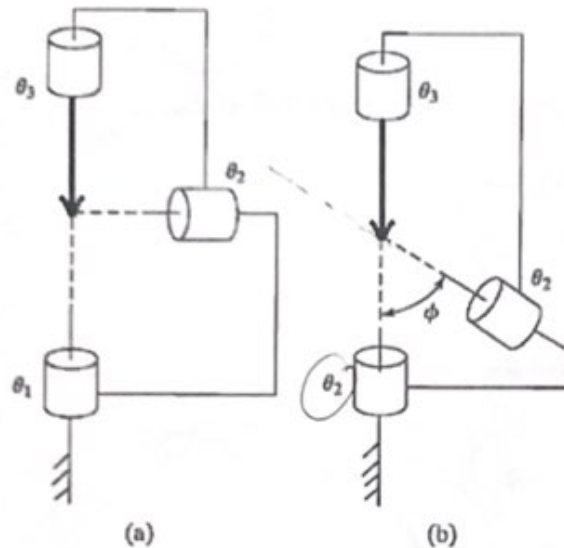
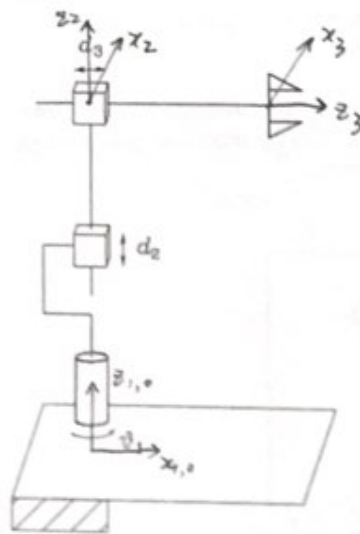


Figure 5: Two 3R mechanisms.

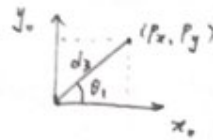


Comprehensive Problem

Given ${}^0_3T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$, attach the frames and solve the inverse kinematics for the cylindrical arm below.



$$d_2 = p_z$$



$$\theta_1 = \text{atan2}(p_x, p_y)$$

$$d_3 = \sqrt{p_x^2 + p_y^2}$$

Figure 6: Cylindrical arm.